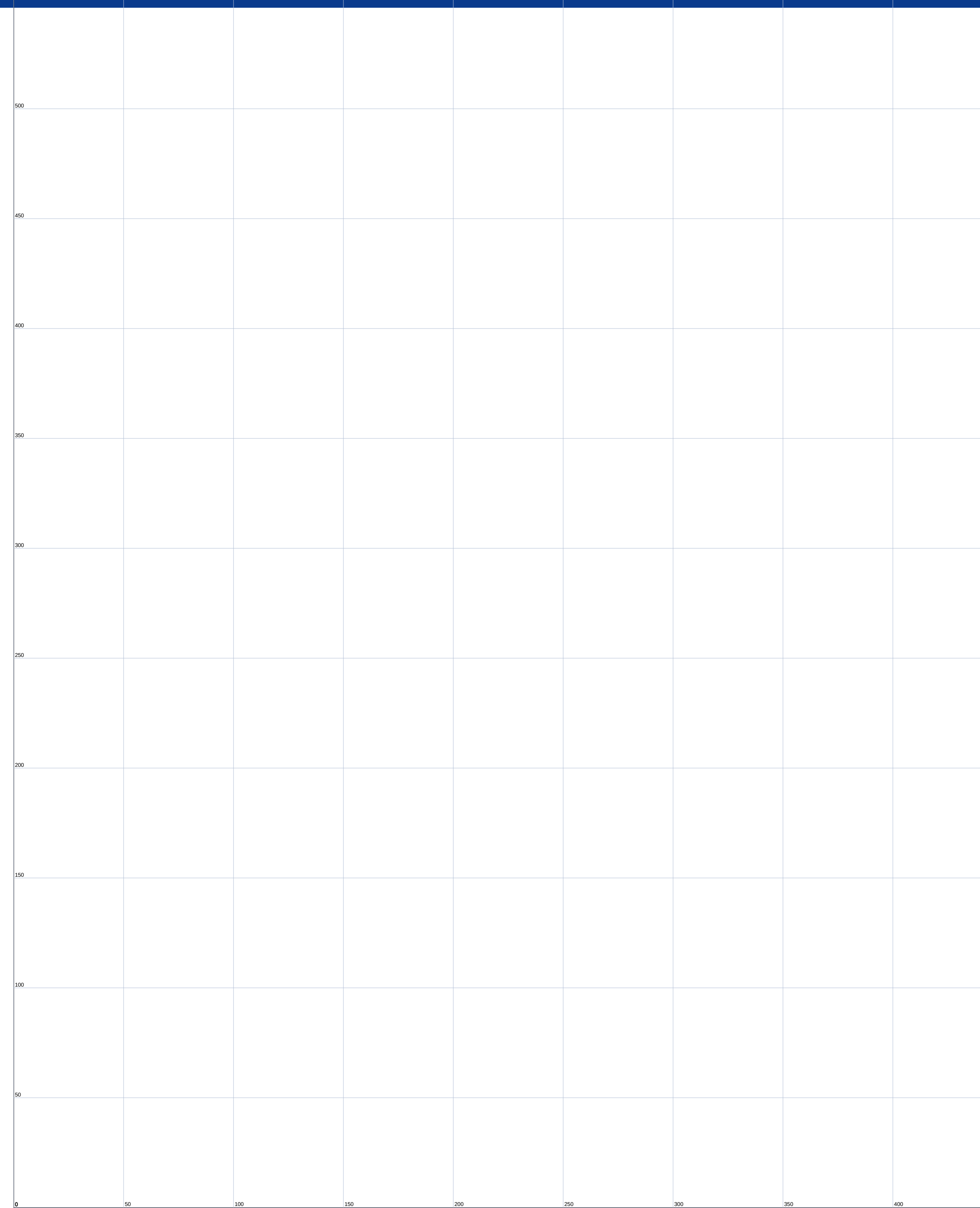


LR-S1A 1U CPO Switch Motherboard — Assembly Drawing (Bot)



Reference-designator summary — Bot side

Prefix	Function	Qty	Examples
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Assembly notes

- 1. SMT assembly per IPC-A-610 Class 3 · 100 % visual inspection.
- 2. Reflow 8-zone Pb-free SAC305 · peak 245 °C ±3 °C · TAL 60 ±10 s.
- 3. LR-CSW-51T2 484-ball BGA · placement ±50 µm · 100 % AXI · 100 % first-article X-ray.
- 4. Underfill LR-CSW composite (Loctite 3593) · corner-bond LR-NCE daughter ICs.
- 5. ESD per ANSI/ESD-S20.20 · HBM 2 kV / CDM 500 V minimum.
- 6. Hand-assembled: ELSFP cages, OSFP-XD cages, MPO inlets, cold-plate, PSU bays.
- 7. Cold-plate retention 4×M2.5 to 0.5 N·m cross pattern · TIM Honeywell PTM7950.
- 8. Fan headers keyed · install fans after PSU bring-up · ≥ 36 mmH₂O at 12 000 rpm.
- 9. PSU bay 25-cycle hot-swap · CRPS pin-mate alignment ±0.3 mm.
- 10. Cleaning aqueous DI + saponifier · ≤ 1.56 µg NaCl/cm² (IPC-TM-650 2.3.25).
- 11. No conformal coating (liquid-cooled, sealed chassis).
- 12. Hi-pot 500 VDC 60 s between V_PSU_12V_A/B and GND · ≤ 1 mA leakage.
- 13. ICT coverage ≥ 95 % nets · functional CPU + BMC + ASIC PLL + HBM init test.
- 14. Date code on B.Silk · IPC-7351B 2D-data-matrix · YYWW format.
- 15. QC: 24 h 65 °C burn-in · full PDN load · fan sweep · 32-port link-up.

LightRail AI — Confidential / Investor data-room

Programme	LR-S1A 1U CPO Switch Motherboard
Drawing No.	LR-S1A-ASM-002
Title	Assembly Drawing — Bot side
Material	18-layer HDI · Megtron-7 / FR-4 / BC24
Designer	LightRail AI — Design Automation
Date / Rev	2026-04-19 · Rev 1.01 / 1